

UTM

UNIVERSITI TEKNOLOGI MALAYSIA

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SECR2213 - 01 NETWORK COMMUNICATIONS

NETWORK DESIGN FOR FACULTY OF COMPUTING

GROUP NAME:
THE COOLIES

PROVIDED FOR:
DR. ADNAN KHALID

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NAURA CAHYA MARISSA DEWI	A25CS4064

1.0 Introduction

The Faculty of Computing, which previously operated from a single storey, has now constructed a new two-storey building, named Block N28B. The primary goal of this project is to develop and implement a network infrastructure that not only meets current requirements but is also adaptable to future technological advancements. The network must be highly flexible to support a wide range of activities, as the building houses diverse functional spaces, including student lounges, classrooms, labs, and specialized facilities such as a Cisco network lab and a video conferencing room.

The network design will ensure scalability to accommodate the faculty's evolving needs. This includes both wired and wireless solutions to provide seamless connectivity for approximately 30 workstations per lab, while also delivering a reliable online environment for both faculty and students.

To initiate the project, we created an architectural floor plan to optimize the placement of network equipment such as switches, routers, and access points, maximizing the efficiency of the infrastructure. We studied best industry practices and evaluated various devices and technologies to select the most suitable equipment for the building's requirements.

A key objective of the network design is to establish a sustainable infrastructure that minimizes the need for frequent upgrades. Our aim is to create a robust, secure, and efficient network capable of supporting the growing academic and administrative population for the next 20 years. By strategically positioning devices, carefully allocating IP addresses, and planning for future expansion, we are laying the foundation for a high-tech academic environment.

2.0 Project Background

The Faculty of Computing (FC) currently has 1800 students (both undergraduate and postgraduate), 100 academic and 40 supporting staff. The Faculty anticipates a 15% increase in both students and academic staff over the next four years, prompting plans to construct a two-storey building to support this growth. The new building will include four laboratories, a video conferencing room, a hybrid classroom, and a student lounge, all equipped with Wi-Fi connectivity.

Each laboratory will have 30 workstations and high-speed internet to meet the educational requirements of the Fourth Industrial Revolution (4IR). The labs include two general-purpose labs, a Cisco Network Lab for training in network implementation, and an Embedded Lab for practical learning in IoT, digital systems, and sensor technologies. Each lab and the student lounge will measure 14m x 10m, providing sufficient space for workstations and equipment in a comfortable learning environment.

To accommodate the growing needs of students and staff, the building's network infrastructure must be scalable, secure, and easily manageable. The Cisco Network Lab requires specialized networking equipment, while the Embedded Lab demands high-performance devices for hands-on learning. The hybrid classroom will be designed to support optimal teaching and learning, and the student lounge will provide a quiet space for relaxation while maintaining network access.

The Faculty aims for a network infrastructure capable of handling any scenario, as emphasized by the Dean. This requires a high-performance, secure, and cost-effective network, resistant to threats such as denial-of-service attacks. Additionally, the network should be expandable to support future wireless connectivity, ensuring that the building can meet both current and future technological demands.

3.0 Compiled Tasks

3.1 Task 1 Project Setup

3.1.1 Floor Plans

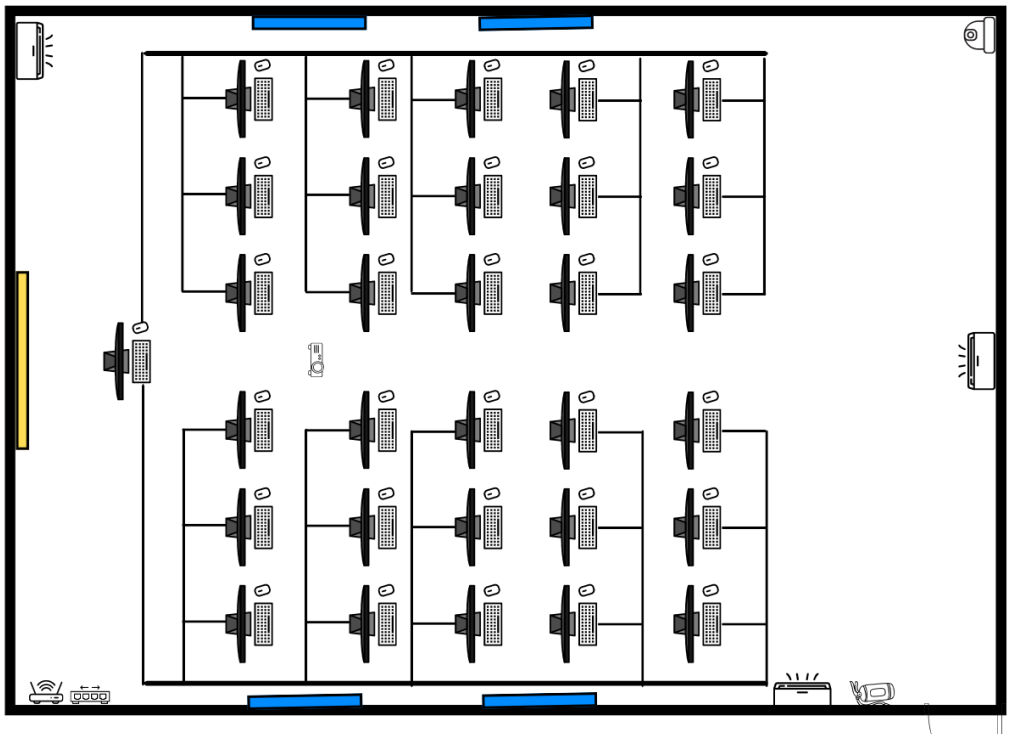


Figure 1 General Lab 1

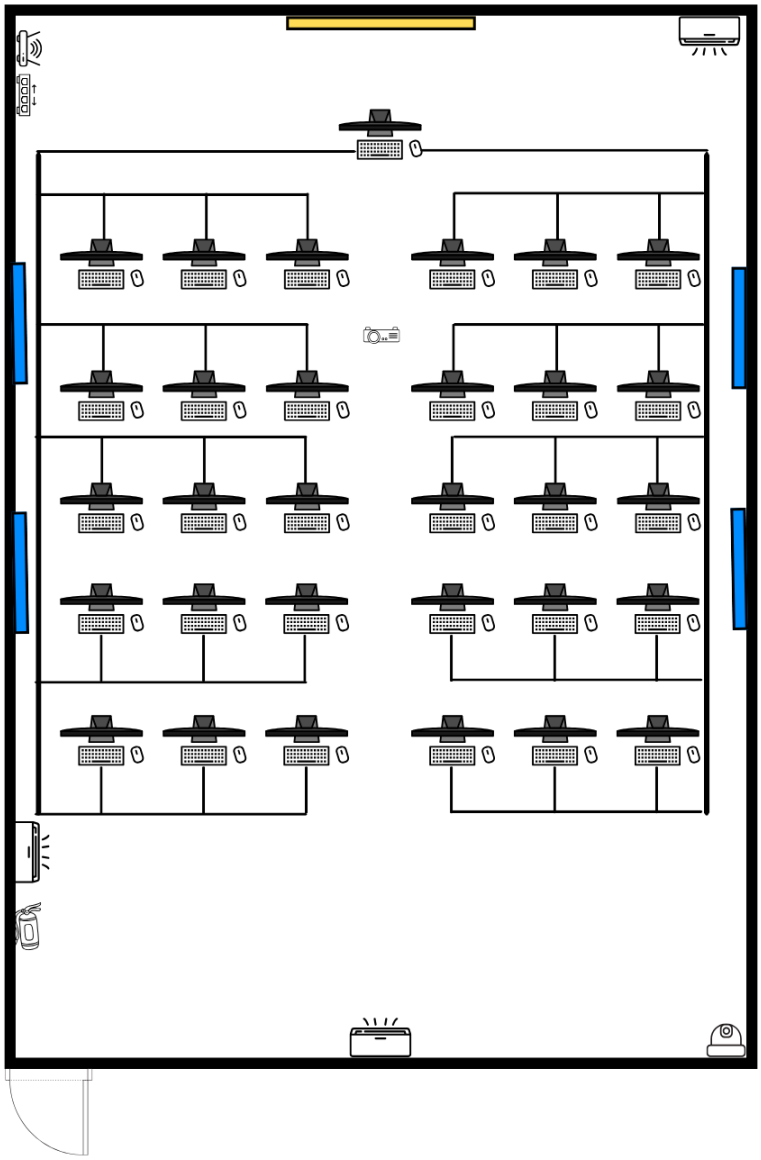


Figure 2 General Lab 2

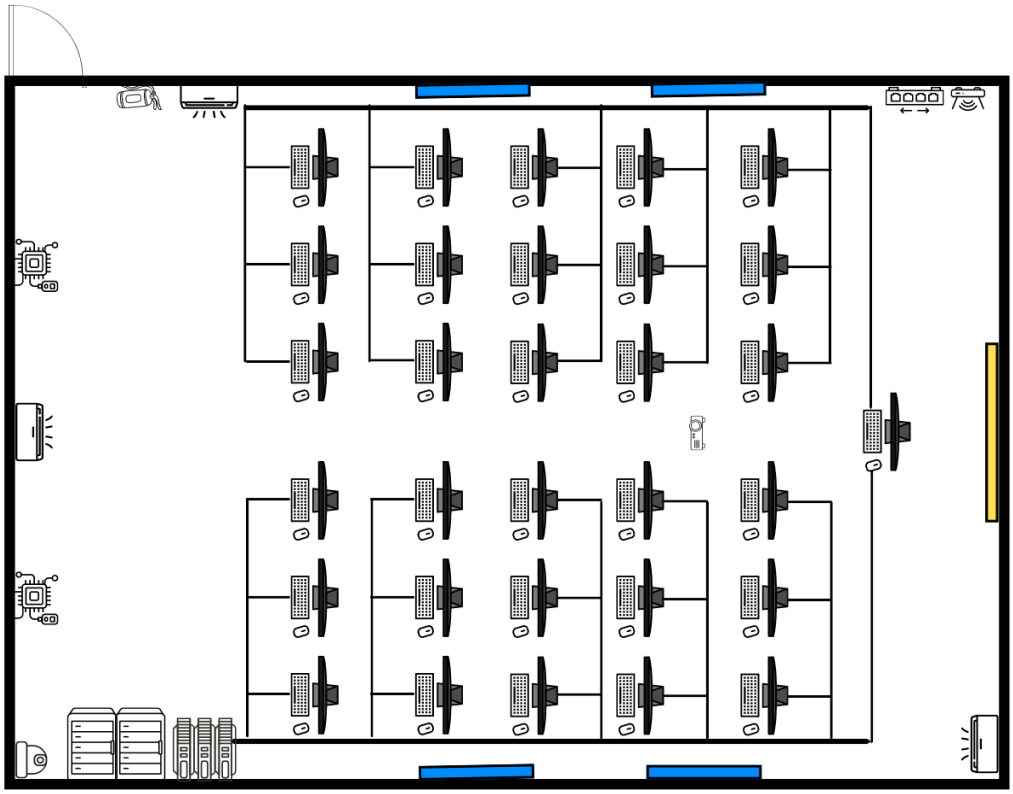


Figure 3 Embedded Lab

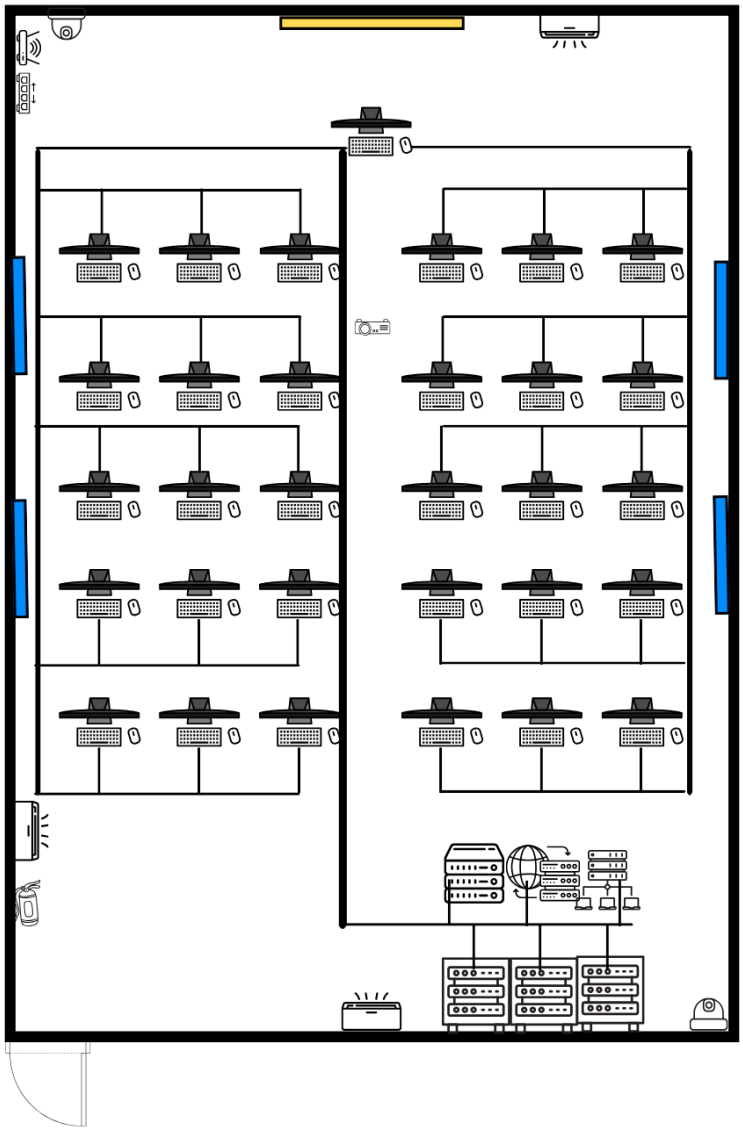


Figure 4 Cisco Network Lab

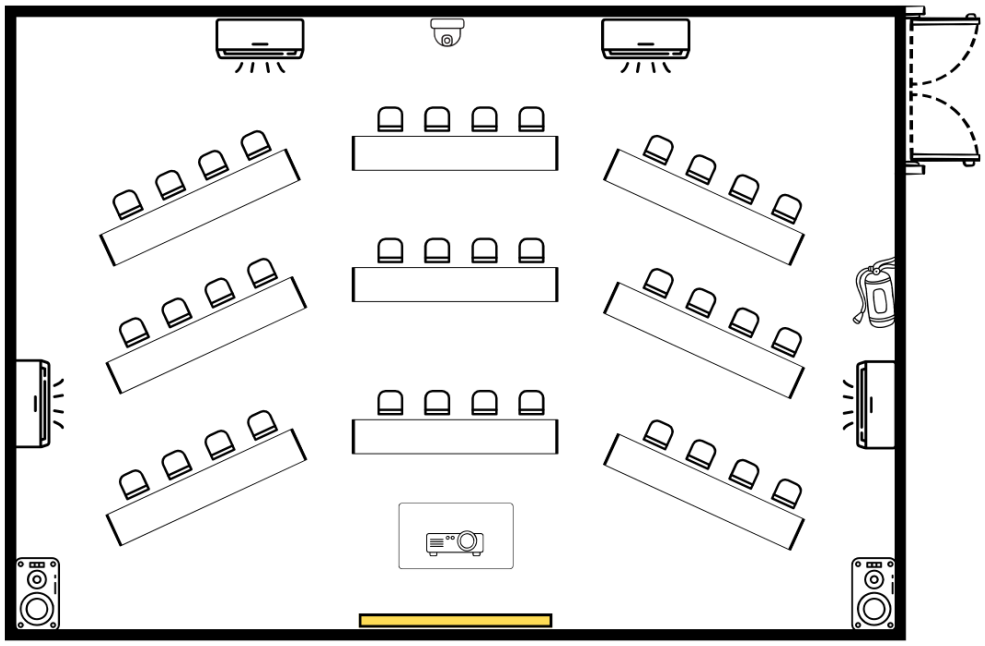


Figure 5 Hybrid Classroom

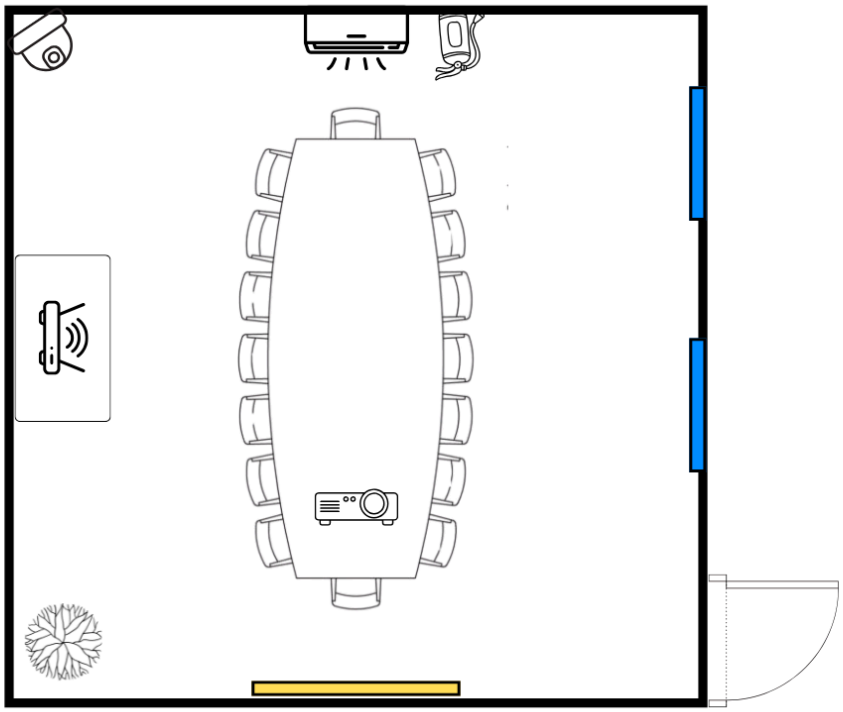


Figure 6 Conference Room

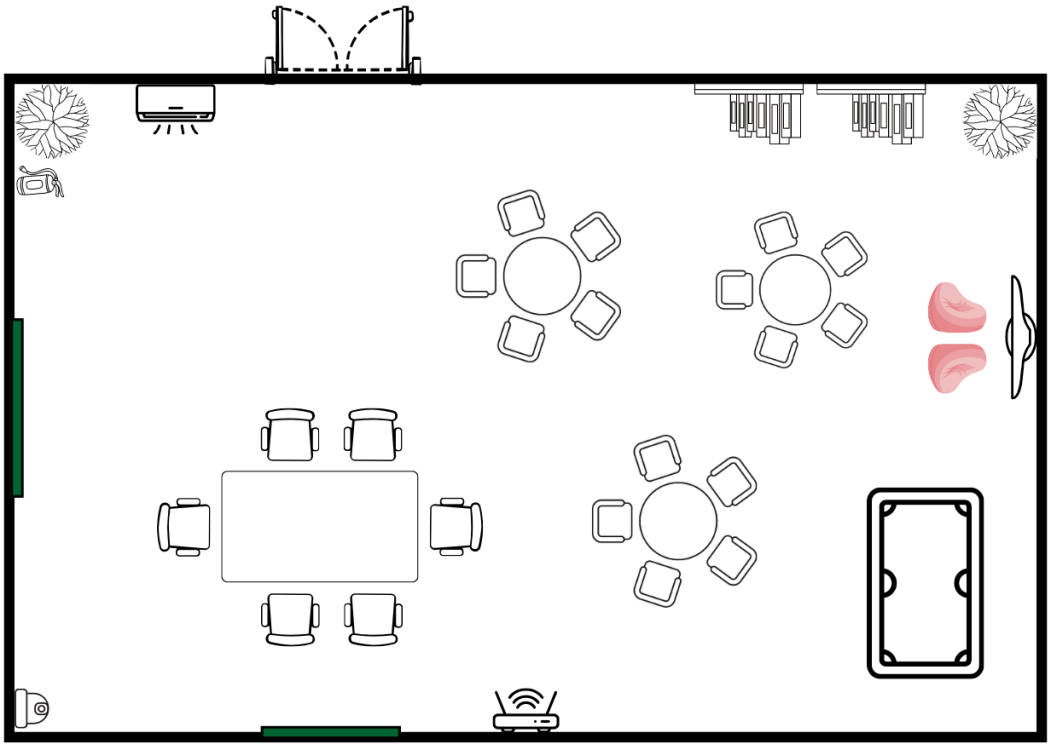


Figure 7 Student Lounge

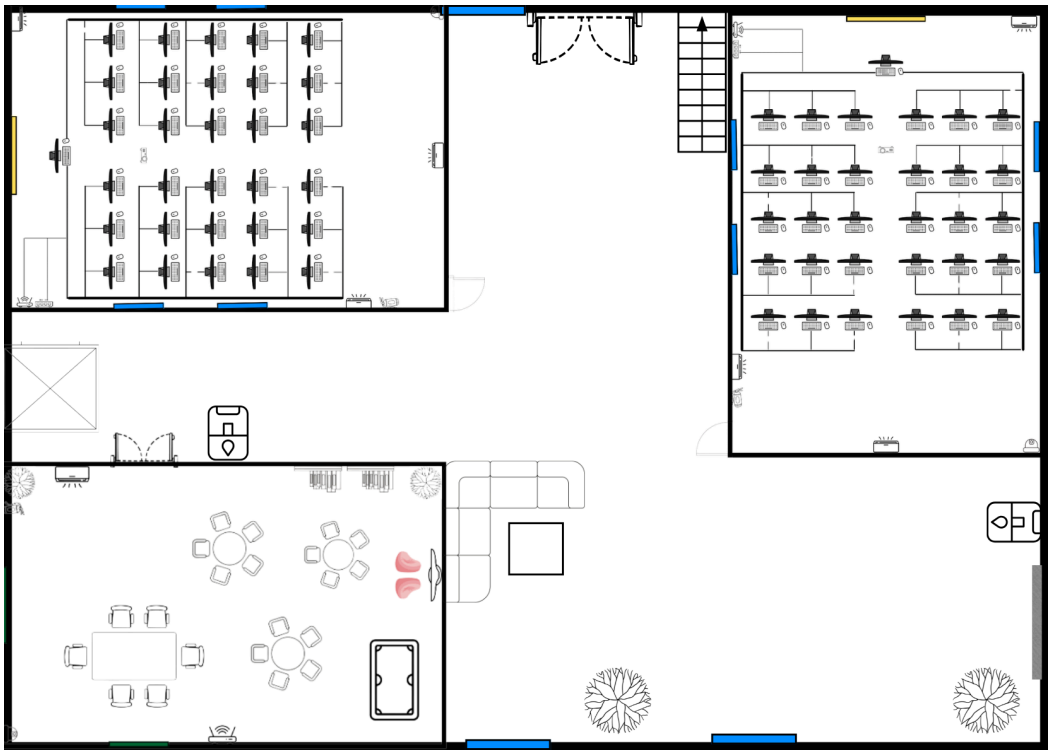


Figure 8 First Floor of N28B

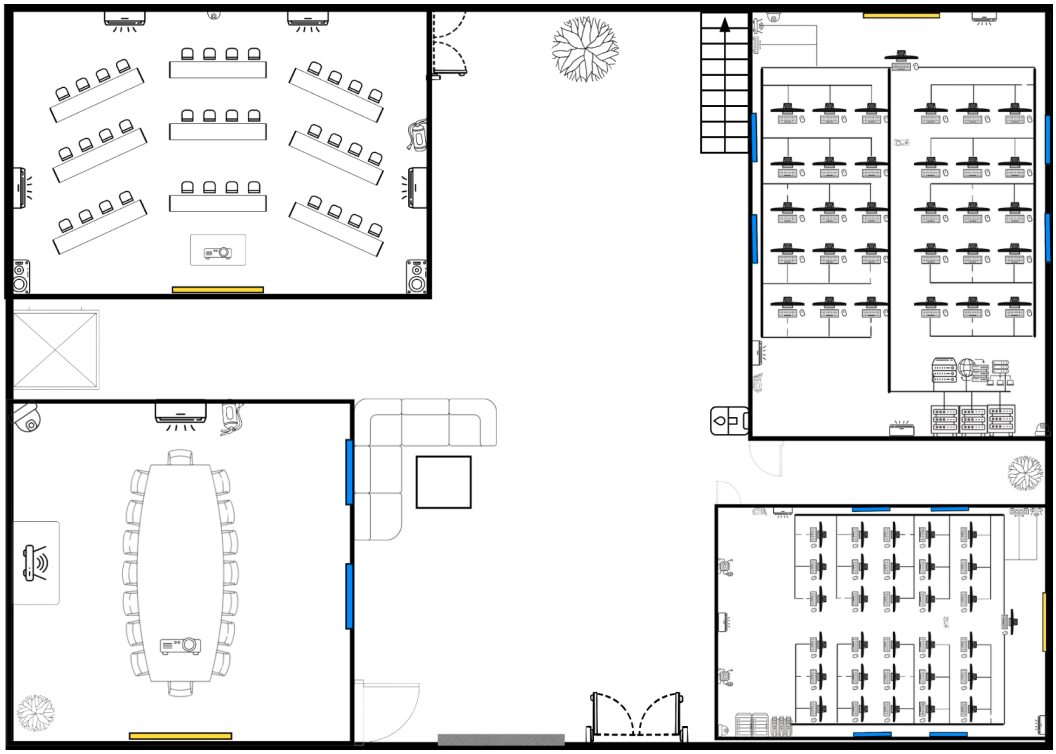
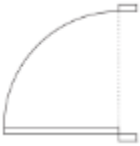
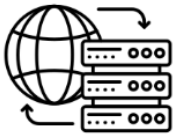
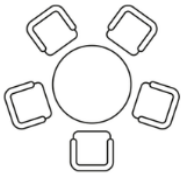
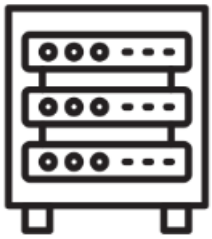
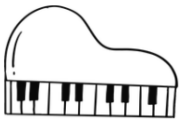
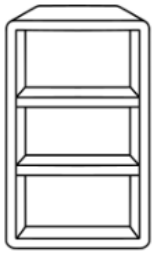
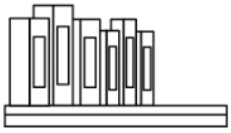
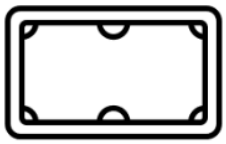
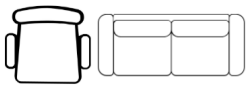
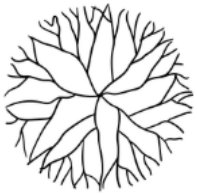
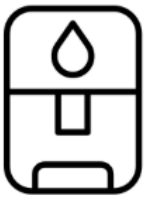
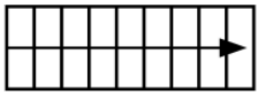
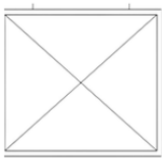
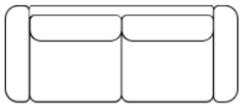


Figure 9 Second Floor of N28B

3.1.2 List of Symbols

Element	Represents	Element	Represents
	Computer		Projector
	Air Conditioner		Door
	Smart Board		Window
	CCTV		Fire Extinguisher
	Wifi Router		Printer
	Switch		Double Door
	Rack for IOT		Speaker

	Server		Proxy Server
	Web Server		Notice Board
	Discussion Table		Rack
	Sensors for Embedded Lab		Piano
	Shelf		Book shelf
	Console		Table

	Billiard Table		Chair
	Bean Bag		Led TV
	Plant		Water Dispenser
	Emergency Stairs		Elevator
	Restroom/Toilet		Sofa

Meeting Minutes

Task 1

DATE/TIME	27/03/2026	
LOCATION	Online Via Google Meet	
AGENDA	Discussing Task 1	
MEETING MC	Adityo	
ATTENDANCE		
NAME	**TIME**	**REASON FOR ABSENCE**
Adityo	2215	
Faustina	2215	
Fabian	2215	
Naura	2215	

MINUTES			
NO.	**ITEM DISCUSSED**	**IDEAS & WHO GAVE IT**	**PERSON IN CHARGE & DATE**
1	Who and where to write the report	In Google Docs (Naura)	Naura (27/03/2026)
2	Who and where to draw the floorplan	Using Canva (Adityo, Faustina)	Faustina (27/03/2026)
3	Who will assist Faustina to draw	Virtually through Canva (Adityo)	Adityo (27/03/2026)
4	Which elements in the floorplans will be used and	By choosing elements in Canva	Fabian (27/03/2026)
5	Next Meeting	04/02/2026	
6	Meeting Ended	2359	

DOCUMENTATION

